

[DC] A Tangible Augmented Reality Programming Learning Environment (TARPLE) for Active, Guided Learning

Dmitry Resnyansky*

School of Information Technology and Mathematical Sciences, University of South Australia, Mawson Lakes, Australia

ABSTRACT

This PhD project aims to bring together technological and educational perspectives by understanding how objectives of programming learning and principles of active learning and embodied learning can be supported and enhanced with AR and TUI technologies. This work presents an approach to the design and evaluation of a TARPLE prototype with enhanced functionality to encourage active guided learning of a text-based OOP language. The system supports natural interaction with learning material, and embodiment and contextualisation of information in 3D space. One of the goals of this thesis has been to understand how empirical studies can inform the design of a TARPLE that supports learning of text-based programming languages and development of a basic debugging skillset.

Keywords: Tangible, augmented reality, programming education.

Index Terms: Human-centered computing—Interaction design; Applied computing—Education—Interactive learning environments

1 INTRODUCTION

This PhD project aims to bring together technological and educational perspectives by understanding how objectives of programming learning and principles of active learning and embodied learning can be supported and enhanced with Augmented Reality (AR) and Tangible User Interface (TUI) technologies. These approaches are combined in a Tangible Augmented Reality Programming Learning Environment (TARPLE).

Based on analysis of previous related work [1, 2], the following research gaps have been established:

- the limited use of hybrid AR TUI-based solutions in tertiary Computer Science (CS) education, in particular for teaching text-based programming languages, for the development of high-order skills such as debugging, and for collaborative learning;
- a lack of augmentation of interface components that provide not only representation of CS and programming concepts but that enhance user interactivity with learning material;
- an emergent need to develop evaluation approaches that bring together observable user behaviours and high-order learning objectives; and
- lack of an integrated framework that could bring together the educational, psychological, communicative, and technological aspects of technology-supported programming learning.

* Dmitry.Resnyansky@mymail.unisa.edu.au

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The expected outcomes of this research include:

- the design of a new generation of educational technology via merging of innovative ARTUI paradigms with theories of active, embodied, and scaffolded learning;
- a methodology for the evaluation of ARTUI-supported programming learning environments;
- empirical exploration of TARPLE uses in a senior secondary school setting and multi-faceted evaluation of its effectiveness for learning retention and impact on the learning process.

This paper reviews work done to date, and outlines future work including empirical user studies to evaluate the experimental prototype and address research questions.

2 CHALLENGES IN DESIGN AND EVALUATION

The present TARPLE design has been guided by two goals: i) Supporting development of programming learning competencies unfolding along two axes: *Interpretation* and *Production* [3]; and ii) Supporting active learning through scaffolding and guiding students' interaction with learning material and by bolstering learner engagement and retention.

These goals and the design approach and solutions to address them were presented in a paper describing an ARTUI-mediated debugging activity paradigm for the teaching of text-based OOP languages [4]. In addition, they were elaborated upon in a paper focusing on immersion through natural interaction for AR-supported anatomy learning [5], as well as in a paper that features an ARTUI design framework that highlights the role of learning scaffolding through feedback, interaction, and collaborative learning for programming instruction [2].

A challenge faced by educators and developers of instructional technology is evaluation of learning outcomes and processes that are mediated by multimodal technology [6]. This PhD study is inspired by the idea that evaluation of innovative educational technologies requires new methods and techniques of analysis.

The goal of empirically measuring action-based behaviours requires medium- and domain-specific methodologies for studying student-user interaction within the system. This is to enhance assessments of both performance and usability with standard methods, such as pre-/post-test scores and the System Usability Survey (SUS). Within research on GUI-based tools for CS education, such constructs and measures have been developed. For example, taxonomies of learner behaviour that indicate different levels of engagement with algorithm animations [7]. However, categories for the analysis of students' interaction with AR and TUI systems in CS education and programming learning are yet to be developed.

This PhD aims to fill this gap by suggesting the types of empirical behaviour that needs to be observed and measured in terms of the AR and TUI effects on learning and the TARPLE usability.

3 PROGRESS TO DATE

So far two prototypes of a TARPLE have been developed, and a mock pilot experiment has been conducted to define evaluation constructs and measures as well as to inform further system

design. A handheld display (HHD) variant of the system was initially developed to provide multiple levels of interaction with learning materials through a simple programming scenario. This variant was tested in an informal study to determine evaluation and design gaps, after which a head mounted display (HMD) variant was developed and is in the final stages of refinement.

The TUI component of the HHD variant comprises a series of physical manipulatives in the form of blocks appended with Vuforia registration patterns for image tracking and augmentation and block names corresponding to the individual code elements of a program statement visible on each block side (except on bottom). The system's AR component comprises marker-based tooltips incorporating two levels of information: a basic level showing a brief explanation of the Java term corresponding to the block name; and a detailed description of the code concept, related syntactical rule, and a textual hint (where applicable) (Fig. 1).

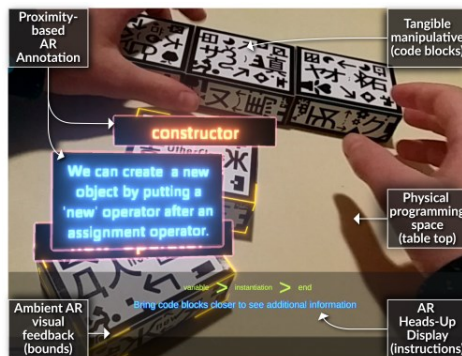


Figure 1: Physical manipulatives both with and without the different levels of virtual cue.

A programming task is performed using the AR TUI by interacting with tangible blocks and observing the AR content in the user's immediate 'learning' workspace (as viewed through the display), and arranging blocks successively within the 'programming' workspace (physical tabletop surface) to construct a statement.

The HHD prototype was tested in an exploratory study on a small sample of users of different age and educational background in order to identify observable actions that may be interpreted as activities to be measured. Four activity types have been identified: *Exploring* (the system); *Learning* (Java terms and their meanings and rules); *Programming* (constructing code statements); and *Verifying* (constructed statement validity). Observations with regard to the system's practical functionality were made and used for informing the design of further prototypes.

Informal user testing using a HHD has helped to identify some worthwhile approaches to the design of an improved system prototype and the elements required to address the technological and pedagogical demands of instructors, including directions for a HMD variant of the environment. This testing has aided the integration of marker-based tracking of physical manipulatives with localised AR visualisation to provide additional user guidance cues.

The system's HMD variant runs on a Microsoft HoloLens and uses similar AR and TUI components. In addition there is a localised AR environment that presents interactive visualisation assets that serve as additional guides for students constructing their program. These assets comprise a virtual code template, glossary, program visualisation, and text cues for providing task context and progress of statement construction.

Initial ARTUI prototypes kept guidance UI features to a minimum, with the aim of freeing users to work and learn

uninhibited, guided simply by the contextual annotation of the TUI. Upon testing the system both with a HHD and a HMD, it became apparent that some users had difficulty even starting the task, and that general augmentation of code blocks alone was insufficient for active learning. Instead, the active learning would be better served by combining it with a guided learning model that required a global AR environment to direct users. The first major change in design was to incorporate a virtual template to approximate a program structure and imply a goal to users.

An interactive code template situated within the user's workspace may help them to take an active role in the construction process, instead of estimating the intended structure of a program statement. Visual cues in the form of a markerless program visualisation and gesture-activated glossary may assist guide users during the construction task.

4 CONCLUSION

The principle novelty of this research is its interdisciplinary position as it strives to bridge technology and innovative pedagogical approaches; implementation of the innovative principle of gamification for grounding the process of coding within broader instructional context to decrease students' reliance on external learning resources; highlighting the potential of HMDs for enriching the AR medium with TUI affordances to enhance embodied learning and spatial cognition.

This PhD study should serve as a guide to take further steps in developing and evaluating more advanced learning modalities such as those involving collaborative interaction.

Procedures for three experimental studies involving both tablet and HMD variants of a prototype AR TUI with measurement instruments for qualitative and quantitative data collection have been elaborated based on prototype testing. Preparation for field studies is currently at the initial stage, and will be followed up by resultant prototype refinement and analysis of data.

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